

Chapter 9, Problem 83P

Problem

An inductive bridge balances when $R_1 = 1.2 \text{ k}\Omega$, $R_2 = 500 \text{ }\Omega$, and $L_s = 250 \text{ mH}$. What is the value of L_x , the inductance of the inductor under test?

Step-by-step solution

Step 1 of 1

$$\begin{aligned} L_x &= \left(\frac{R_2}{R_1} \right) L_s \\ &= \left(\frac{500}{1200} \right) (250 \times 10^{-3}) = 104.17 \text{ mH} . \end{aligned}$$